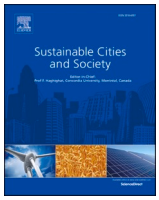
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Participatory Framework for Urban Pluvial Flood Modeling in the Digital Twin Era

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ABSTRACT

The recent advancement in digital twin technology, which creates virtual replicas of real-world processes, offers an interactive testbed for understanding and predicting environmental changes. As pluvial flood damage escalates globally in urban areas, there remains a gap in understanding the most effective collaboration between governments and local residents for sustainable flood risk management. To address this gap, we develop a participatory framework for urban pluvial flood modeling, incorporating open source software, virtual reality, minimum viable product, and gamification components. This framework engages citizens in every phase of the participatory modeling process, from input data preparation, through hydrological model construction, to model verification, and experiments. We present a case study on the recurring pluvial flood damages in South Korea's Gangnam region, demonstrating the practical implications of an interactive, web-based crowdsourcing platform to leverage community engagement and local knowledge. The results underscore the evidence of a proactive role for citizens, not merely as recipients of disaster information but as key contributors collaborating with a range of stakeholders in stormwater management and modeling. Combining digital twin technology with citizen participation can empower informed decision-making and collective actions in the evolving digital era, leading to a disaster-resilient community.

1. Introduction

A Digital Twin, defined as a virtual replica of physical entities, processes, and systems in the real world, has burgeoned with the rapid advancement of environmental monitoring and computing power over the last decade (Batty, 2018). The digital twin allows us to visualize, analyze, and simulate how the physical counterpart would behave by assimilating the physical and digital world (Tao et al., 2019). One of the best ways to leverage this technology is to predict future environmental changes and to explore optimal interventions (i.e., new designs, technologies, or policies) in advance (Fuller et al., 2020). This concept is widely accepted in areas ranging from manufactured products to buildings, cities, and even ecosystems (Qi et al., 2021). Recently, there have been many attempts to employ digital twins at the city scale to assess adaptive strategies for climate change-induced disasters, especially in areas where actual social experiments are not feasible due to economic and moral concerns (Weil et al., 2023; Xia et al., 2022). In the field of stormwater management, especially with the potential for significant flood risk reduction through government-citizen collaboration, the call for implementing digital twin platforms is on the rise (Weil et al., 2023).

As extreme heavy rainfall events increase due to climate change, not only the flood-prone areas along rivers but also inland urban areas are facing the risk of pluvial flooding, which occurs when the intensity of rainfall exceeds the capacity of urban drainage systems (Lin et al., 2022; Rosenzweig et al., 2018). Urban pluvial flood modeling has been conducted primarily from the supply side, involving entities such as the government, academia, and industry (Azizi et al., 2022). This top-down approach has mainly focused on advancing the detailed methodology, spatial and temporal resolution, and accuracy of the hydrological model

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(Bulti & Abebe, 2020; Lim et al., 2023). While this conventional modeling approach was suitable in the past when data was generated and utilized by a few entities, its high cost and 'black-box model' nature—where citizens only receive the result of flood risk map without understanding the underlying mechanisms—hinder social support and consensus (Gebremedhin et al., 2020). Recently, there has been an emphasis on involving citizens directly in urban pluvial flood modeling from the demand side of flood control (Assumpção et al., 2018). This bottom-up approach uses participatory modeling to socialize the costs, allowing consecutive model updates in the public domain and validation using volunteered geographic information (VGI), like social media images (Haworth, 2016; Lin, 2022; Vermiglio et al., 2022). The participatory modeling approach can be effective in enhancing the granularity of large-scale urban modeling (Zhang et al., 2019).

Pluvial floods can occur not only in low-lying areas but also in any urban area experiencing structural or functional failures of the drainage system, whereas fluvial floods are mainly associated with specific flood-prone areas (Douglas et al., 2010; Lin et al., 2021). This imposes greater spatial uncertainty on pluvial flood risk, with the risk becoming even higher in densely populated areas that have undergone rapid urbanization (Di Salvo et al., 2018; Lin et al., 2023). Therefore, while it is necessary to construct pluvial flood modeling across the entire urban area, the high cost of a top-down approach underscores the need for citizen participation (Assumpção et al., 2018). By engaging the participation of local residents, it is possible to incorporate diverse local knowledge into the modeling, such as historical changes in the hydrological regime, water courses, flood patterns, and socio-demographics (Haworth, 2016; Hendricks et al., 2018; Wehn et al., 2015). Enhanced citizen ownership and transparency in participatory modeling lead to community engagement, empowering individuals to have a voice in decision-making and thereby promoting accountable and sustainable disaster management (Gray et al., 2018; Zhang et al., 2019).

Urban pluvial flooding, which encompasses both surface runoff and combined sewer overflows, represents a challenge intertwined with complex natural and human systems (Liu et al., 2007). In light of this, there has been an increasing call for a paradigm shift towards participatory stormwater management, emphasizing collective actions (Bouzguenda et al., 2019; Dhakal & Chevalier, 2016; Lim et al., 2023). By taking measures such as sand-bagging, installing temporary water barriers, and cleaning inlets (e.g., drain grates and curbs) or trenches, citizens can play an instrumental role in mitigating pluvial floods (René et al., 2018). As such, a bottom-up approach to stormwater management is pivotal, where citizens are actively involved in every phase of the process, from building pluvial flood models to analysis, decision-making, evaluation, and feedback (Barreteau, Bots, & Daniell, 2010; White et al., 2021). By doing so, citizens are encouraged to brainstorm and implement preventive measures at both the individual and community levels (Lehtola et al., 2022). Consequently, a platform that provides prompt feedback on the anticipated effects of those actions is vital. However, there is a lack of comprehensive and user-friendly guidelines for participatory pluvial flood modeling, while each necessary technical component has already been developed as open-source (White et al., 2023). To address the gap, we propose a framework for interactive and public-driven rapid urban pluvial flood modeling, along with a step-by-step case study that walks through the entire process.

This paper discusses four key components essential for participatory urban pluvial flood modeling: Open Source Software (OSS), Virtual Reality (VR), Minimum Viable Product (MVP), and Gamification (Fig. 2). First, OSS liberalizes access to urban flood modeling by eliminating software costs and enabling seamless integration with different OSS tools (White et al., 2023). OSS platforms enable continuous model improvement, making them more responsive to public needs (Zhang et al., 2022). Second, VR is identified as a pivotal tool for effective communication, providing immersive and realistic visualizations (Portman et al., 2015). VR enhances situational awareness of urban flooding, allowing laypersons to explore the expected impact of floods in various scenarios (Bakhtiari et al., 2023; Fan et al., 2020). Third, by adopting an MVP approach, the web-based participatory model aims to accelerate computational speed while still capturing essential physical processes, considering both budget and data constraints (Bartos & Kerkez, 2021). This parsimonious approach can enable near real-time flood assessments and early warning systems, keeping the platform versatile and straightforward (Sermet et al., 2020). Fourth, gamification incorporates the behavioral aspects of citizens to encourage public participation (Baruch et al., 2016). Even a non-monetary reward system can enhance both the quality and quantity of crowdsourced data (R I Ogie et al., 2019). This integrates well with a sanctions system to deter the spread of incorrect information and facilitates citizens acting as both users and providers of data (Ostrom, 1990). Collectively, these four components work synergistically to establish a robust, participatory modeling to foster grassroots initiatives for sustainable stormwater management in the digital twin era.

We provide a framework for participatory urban pluvial flood modeling that involves a multi-step process, starting with preparing input data for constructing and calibrating hydrological models. The framework couples a one-dimensional stormwater model with a rapid flood spreading model (RFSM) to offer a simplified yet effective tool for rapid flood risk estimation (Jamali et al., 2018). Model verification relies on a range of data sources, such as flood inundation maps, flood marks, sewer level data, social media images, and anecdotal reports from local residents (Fohringer et al., 2015; Le Coz et al., 2016; Wang et al., 2018). Additionally, the framework is tailored to explore various flood mitigation strategies, including both conventional 'grey' infrastructure solutions like retrofitting drainage systems, as well as 'green' infrastructure like installing various low impact development practices (Hendricks & Dowtin, 2023). The framework also incorporates the role of community engagement and future climate change scenarios to explore solutions for contributing to urban flood resilience and climate change adaptation. We demonstrate this framework with a case study on recurring pluvial flooding in the Gangnam region of Seoul, South Korea, using popular open-source software like EPA SWMM and QGIS. The framework aims to provide a participatory and process-based methodology for developing near real-time urban flood models, offering actionable strategies and immediate feedback to support various stakeholders, including local residents, engineers, researchers, and policymakers (Geertman & Stillwell, 2004).

The remainder of this paper unfolds in the following order. Section 2 provides an overview of the theoretical background of the four components for participatory urban pluvial flood modeling. In Section 3, we outline the framework for modeling, which includes input data preparation, hydrological model construction, model verification, and model experiments. In Section 4, we showcase detailed guidelines on essential techniques for each modeling framework, only without recruiting actual human participants from the study area. This case study utilizes open source software through every modeling process. In the concluding sections, we highlight the primary insights and implications of our framework in the era of the digital twin. Before we begin, note that the purpose of our model is not to develop and publish the actual working repository or archive for the participatory urban pluvial flood modeling. Rather, we aim to enable general readers to construct and analyze interactive, web-based 3D rapid urban pluvial flood modeling from scratch, following the framework without using any proprietary tools. Our work can contribute to developing a digital twin platform and mobilizing citizen participation around the globe for adapting to urban pluvial flooding intensified by climate change. Consequently, this paper advocates for a shift in perspective, where citizens can play a significant role beyond merely receiving emergency information, by actively participating in data provision, model development and verification, and decision-making processes.

2. Components for participatory urban flood modeling and digital twins

Participatory modeling utilizing digital twins can be conceptualized as revolving around two interconnected axes: the technical aspect and the social aspect (Fig. 1). In the technical aspect, the Level of Detail (LoD) is primarily used to represent the granularity and complexity of virtual reality in relation to its physical counterpart (Lehtola et al., 2022). The LoD is usually divided into 4 to 5 levels (i.e., LoD 0-4). The higher the level of detail, the more accurate and precise the simulation of reality, making the highest LoD ultimately the desired target of a digital twin. However, building a higher LoD requires an extensive amount of input data and often specialized proprietary software, hence significantly boosting the computing power required for simulations (Kikuchi et al., 2022).

From the corresponding social aspect, we propose citizen participation as a solution that can effectively address the technical limitations of a digital twin. Voluntary citizen participation can yield a large amount of detailed and localized input data by socializing the cost of data acquisition (Hendricks et al., 2018). By using open-source software, we can significantly reduce modeling costs by involving citizens throughout the entire modeling process—including model construction, verification, and experiments. Additionally, making data and decision-making open to the public throughout all stages of the modeling processes can bolster the model's reliability and transparency. However, as the required LoD of the model increases, the task difficulty may become too high to expect voluntary participation from the general public. Therefore, within this trade-off relationship, striking an appropriate balance between the technical aspects of the digital twin and the social aspects of citizen participation is crucial for optimal participatory modeling (Rodríguez-Bolívar et al., 2018). To address this challenge, we break down participatory modeling into four components and their relevant attributes within the context of urban flood modeling (Fig. 2).

2.1. Open source software

To encourage citizen participation in urban pluvial flood modeling, the use of open source software (OSS) is imperative. One of the major obstacles impeding active public involvement in participatory modeling is the high licensing costs of specialized hydrology software (White et al., 2023). Recently, OSS has become virtually indistinguishable from proprietary software in terms of functionality, and its user-friendliness has also significantly improved (Namgyal et al., 2023). Additionally, the rapid expansion of online communities among OSS users is creating a virtuous cycle that further promotes the development of OSS. Thus, OSS is a public-driven, flexible platform that permits continuous model refinement and adaptive modifications in response to public demand without relying on a limited number of developer groups (Zhang et al., 2022).

Above all, there is the advantage of improving interoperability through integration with different OSS. Specifically, in the realm of urban flood models, the coupling of stormwater models with overland flow models is essential, and OSS can facilitate this process more seamlessly (Sadler et al., 2019). The scalability of such OSS allows it to be easily transplanted and customized in areas with restricted budgets to build their digital twins, or in countries vulnerable to climate disasters (Nkwunonwo et al., 2020). This contributes to improving environmental justice and equity by opening access to the source code and data of flood models to the public (Dixon et al., 2021; Hendricks et al., 2018). Also, OSS offers transparency in its architecture and algorithms, which can be openly audited and validated by anyone. Thus, this transparency can increase reliability in the modeling product and enhance the sense of community and trust among users, thereby aiding in the improvement of social capital for a more resilient disaster response (Aldrich & Meyer, 2015; Hasala, Supak, & Rivers, 2020; Weil et al., 2023).

As diverse stakeholder groups (including local community members, civil societies, researchers, and policymakers) become more engaged and familiar with these open-source tools, they can enhance the social consensus on emergency responses and collective actions related to the risk of pluvial flooding (R I Ogie et al., 2019). In other words, one person's output can be used as another person's input, creating a linked model chain and crowdsourcing platform. Such collective action and intelligence from the grassroots level are fundamental for proactive stormwater management and pluvial flood mitigation.

2.2. Virtual reality

Virtual reality (VR) is a multifaceted technology which can facilitate realistic visualization and also work as an effective means of communication across various domains (Portman et al., 2015). Ordinary people may find it difficult to comprehensively assess the situation by looking at a traditional 2D flood map with cross-sections representing flood depth (Bakhtiari et al., 2023). Accordingly, VR is a key technology that can improve the situational awareness of the general public in disaster situations by providing an immersive experience similar to the real world in a digital twin (Fan et al., 2020). Additionally, experts from various fields such as urban planning, hydrology, transportation engineering, law enforcement, architecture, and policy making, can virtually explore different flood scenarios. By doing so, they can compare the expected results promptly, allowing them to identify the best strategies through iterative scenario analysis and parametric studies.

Recently, many platforms have been developed that integrate geographic information systems (GIS) with VR to visualize analysis results in real-time (Tobias et al., 2016). The dynamic representation of flood inundation over time facilitates the development and evaluation of emergency response strategies for public safety, including flood protection, evacuation processes, and traffic control (Wolf et al., 2022). For instance, VR simulation of flood inundation extent and depth can

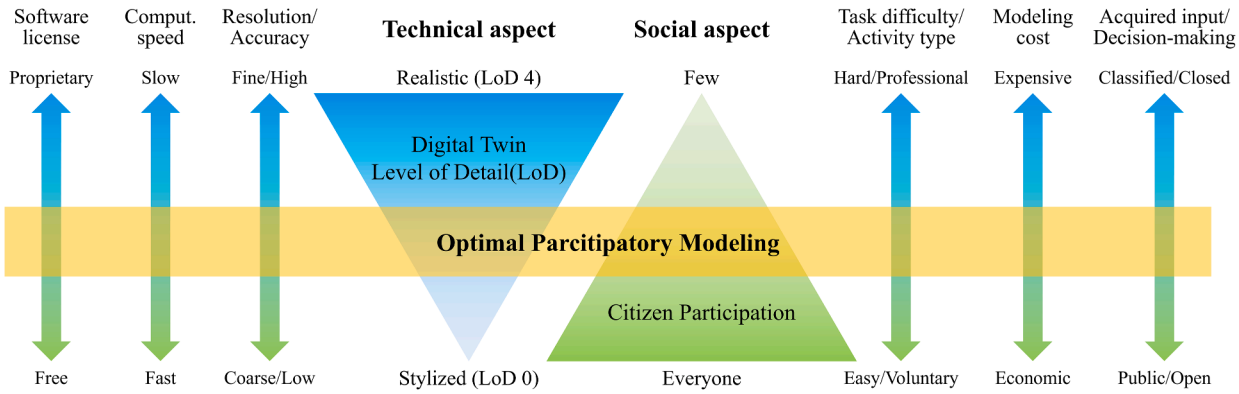


Fig. 1. Trade-off relationship between digital twin technology and the scale of citizen participation for optimal participatory modeling.

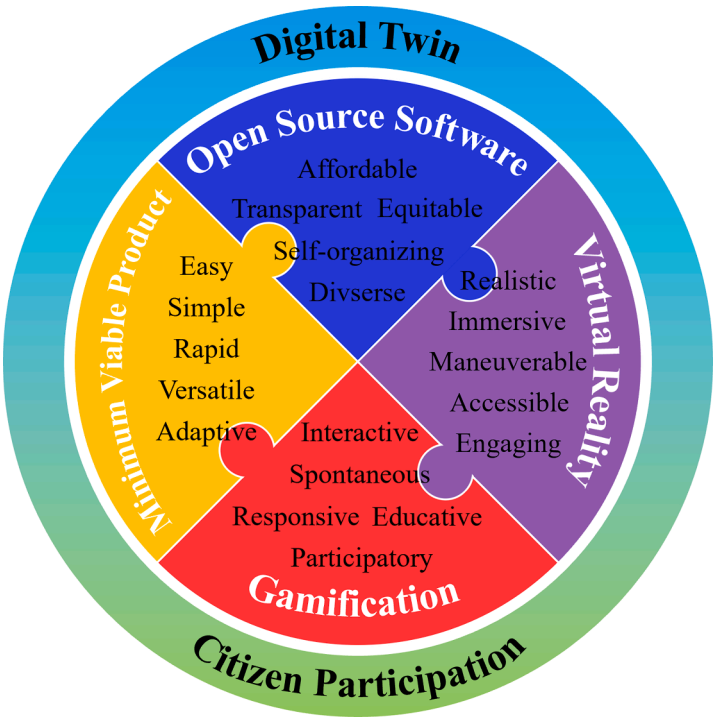


Fig. 2. Four components for participatory urban pluvial flood modeling and their relevant attributes.

provide a basis for determining the height or length of temporary water barriers or sand-bagging for flood damage prevention (White et al., 2021). Furthermore, by offering users the ability to actively navigate simulated flood areas, VR allows them to personalize the potential risks they may face, thereby encouraging individual disaster preparedness (Amirebrahimi et al., 2016).

Particularly for the younger generation, VR can serve as an engaging and informative tool (Luo et al., 2022). Not only are they among the most vulnerable groups at risk of pluvial flooding, but they are also adept at using virtual environments through digital devices and technologies such as the metaverse and the Internet of Things (Rodríguez Bolívar et al., 2023). Thus, VR helps to lower barriers to entry and makes it more accessible for them. By the way, gear equipped with gyroscopes (e.g., VR headsets) for fully immersive VR has not yet been widely disseminated to the general public, the desired detail of this web-based participatory modeling framework is to construct a semi- or non-immersive VR environment (Chassin et al., 2022). For maneuverability, users can navigate a 3D visualization of vertical depth using a keyboard and mouse on a typical desktop PC or their fingers on mobile devices (Kolagani & Ramu, 2017).

2.3. Minimum viable product

The participatory modeling essentially needs to be a web-based platform accessible to everyone at all times (Ma et al., 2024; Wolf et al., 2022). Therefore, it is necessary to ensure sufficiently fast computational processing under the technical constraints of web-hosting to maintain reliable services, especially during emergencies (Bartos & Kerkez, 2021; Voinov et al., 2018). Prior research suggests that the public's average patience for waiting for a website to respond is only 4-8 seconds (Galletta et al., 2004), implying that a rapid response is essential for active citizen participation. The urban flood model typically takes longer in computational speed because it requires the integration of both the stormwater model and the overland flow model (Namgyal et al., 2023; Sadler et al., 2019). The computation speed varies widely for both the stormwater model (depending on the drainage network complexity) and the overland flow model (depending on the dimension of flood modeling methods: rapid flood spreading, 1D, 2D), but it roughly ranges from a few minutes to several hours (Bulti & Abebe, 2020).

If an urban flood model takes longer than, say, tens of minutes for a single simulation, it would be challenging for the platform to actually be used by the general public. Given the nature of disaster information platforms that do not yield revenue, there are budget constraints on expanding the server's hardware capacity. Therefore, every possible method to increase computational speed should be utilized as much as possible on the software side. For example, there are methods such as using lower dimensions of flood modeling methods and employing predefined flood volume-depth curves for the target area to drastically reduce computation time (Rahman et al., 2016).

To get out of a black box model, merely sharing the open-source code transparently (as discussed in Section 2.1) is not sufficient. If the model is too specialized and intricate, and only a few experts can understand it, it will still remain a black box model to the general public. Therefore, for participatory modeling, it is essential to adopt a minimally viable model that is intuitively understandable by the general public while still stylizing and encapsulating the core physical processes (Zhang et al., 2022). By being stylized, the model can prioritize the influential factors affecting urban pluvial flooding, allowing the general public to focus on primary concerns without getting overwhelmed by minute details. Therefore, from the standpoint of the general public, a framework is desired to produce a minimum viable product which is easy and light, suitable for real-time rapid flood assessment, early warning, and emergency operations (Sermet et al., 2020).

2.4. Gamifications

For spontaneous participation from citizens, it is essential to have a two-way feedback platform that facilitates interaction between citizens, as they are both users and providers of participatory modeling (Luo et al., 2022). For instance, a system should inform citizens about anticipated pluvial flood risks in specific areas and allows them to contribute observational data for model validation, effectively acting as a bilateral crowdsourcing platform.

Given the nature of urban pluvial floods, which can emerge and disappear in a short time in unspecified areas, there is often a lack of data on inundation areas or depth (Manfreda et al., 2014). Volunteered geographic information (VGI) can provide near real-time data on flood inundation extent, depth, and flow directions (Cariolet et al., 2019; Yao & Wang, 2020). Compared to traditional physical or remote sensing methods, VGI offers a wider spatial coverage with a high-resolution (Li et al., 2018). Additionally, for the same period and location, on-the-ground observations from multiple participants allow for cross-checking (Y. Li, Osei, Hu, & Stein, 2023). Flood marks or flood scene images can be used to validate the results of the pluvial flood model and enhance its accuracy (René et al., 2018). Furthermore, non-imagery crowdsourced data such as anecdotal evidence and historical flood patterns can be qualitatively analyzed (Rollason et al., 2018).

However, verifying the reliability of VGI and ensuring data security and privacy are necessary. A significant problem in utilizing VGI for emergency management is the propagation of incorrect information, whether maliciously or by mistake (Wang et al., 2018). During actual flood disasters, there are instances where flood scene images are incorrectly geo-tagged, so it is preferable to have photos that also show surrounding geographical features for reference. Moreover, there is also an issue where past flood scene images from the same area are redistributed, causing social confusion. In such cases, it is necessary to determine whether the photo is the same as one previously published by filtering using the photo's metadata (Yao & Wang, 2020). VGI may inadvertently expose sensitive information, such as locations, demographics, or identities (Li et al., 2023). These security and privacy vulnerabilities necessitate robust measures, including personal information protection, anonymity protocols, and clear privacy policies, to ensure the trustworthiness and ethical use of VGI (Yan et al., 2024). Balancing the quality of VGI with privacy and security issues is crucial for maintaining responsive and sustainable public participation (Rodríguez-Bolívar et al., 2018).

To facilitate spontaneous citizen participation, a reward system for those providing information is vital. The reward does not necessarily have to be a monetary payoff; it could be non-monetary payoffs like reputation, social connections (e.g., number of followers), virtual goods, or influences, which give a sense of warm glow or accomplishment (Brown et al., 2016; Ogie et al., 2019). For the reward system to work effectively, a personalized virtual identity that can receive credit is preferred over a completely anonymous identification (Daniel et al., 2018). When people receive immediate rewards for their actions, they become more motivated to participate. This concept, known as gamification, is widely adopted in behavioral experiments and crowdsourcing platforms (Baruch et al., 2016). Simultaneously, sanctions are also necessary for users who continuously provide incorrect information, to prevent 'the tragedy of the commons' in the virtual world as well (Ogie et al., 2017; Ostrom, 1990). Therefore, by leveraging gamification for educational purposes, we can empower voluntary and responsible citizen engagement, ultimately achieving sustainable participatory modeling through a learning-by-doing approach.

3. Participatory framework for urban pluvial flood modeling

3.1. Input data preparation

As the first step, the preparation of various input data is necessary for participatory urban pluvial flood modeling (Fig. 3). The input data can be broadly categorized into data for both model construction and model calibration and validation. For hydrological model construction, input data such as rainfall data, terrain data, building data, land use and land cover (LULC), soil data, and urban drainage network map are needed at a minimum.

In terms of terrain data, not only the digital terrain model (DTM) is needed, but also a digital surface model (DSM) that takes into account the heights of buildings is required for more accurate overland flow modeling (Huang et al., 2024). Using GIS, one can calculate the average slope according to the desired spatial scale (Lin & Li, 2019). For rainfall data, continuous precipitation data from within or near the target area is needed. Given the characteristics of urban pluvial floods, which have short times of concentration and flood durations, a rainfall monitoring

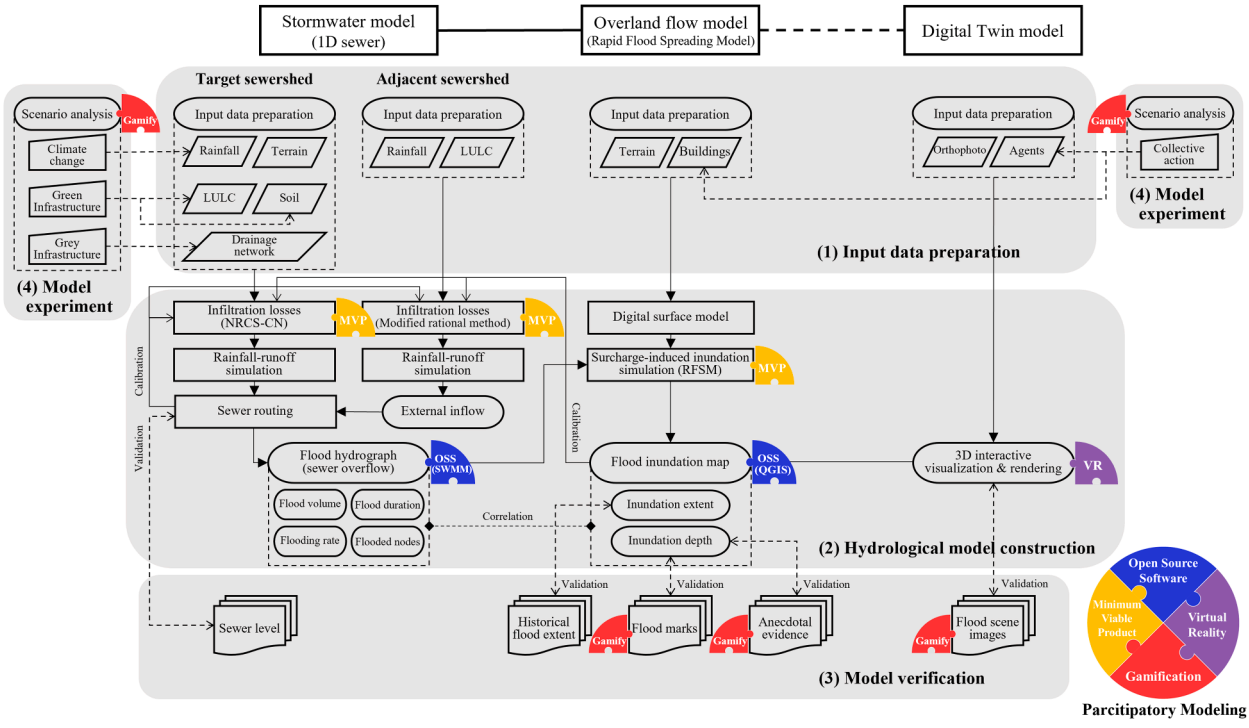


Fig. 3. A participatory framework for urban pluvial flood modeling that encompasses the feedback process from input data preparation, through hydrological model construction, to model verification and experimentation, with respect to the four components of participatory modeling.

interval of ideally less than 20 minutes is preferred (Blanc et al., 2012). Nonetheless, to minimize computational demands and given that the typical precipitation monitoring interval is hourly, the temporal resolution of rainfall data for pluvial flood simulation should be at least hourly (Blanc et al., 2012).

To estimate surface runoff, we chose an empirical and data-driven method: the Natural Resources Conservation Service (formerly called as Soil Conservation Service) Curve Number (NRCS-CN) method (USDA-SCS, 1986). This is because the NRCS-CN method requires less data, specifically LULC and hydrologic soil group data, compared to process-based rainfall-runoff models that demand high-resolution and extensive data. Therefore, we considered the NRCS-CN method more suitable for participatory modeling, as it facilitates greater public engagement through its simplicity and minimum data requirements. With the aforementioned data, a generic rainfall-runoff model can be constructed.

However, to adequately capture the characteristics of urban pluvial floods that occur due to insufficient capacity of the urban drainage system, geometrical data on the urban drainage system for stormwater modeling is needed. Specifically, geographical information about the location or depth of manholes, drain grates, and outlets, as well as conduit diameter and length, is required (Dai et al., 2018; Mohebbi et al., 2020). In most cities, this information is restricted from public access and is classified as confidential data (Blumensaat et al., 2012; Dai et al., 2020; Du et al., 2019; Li et al., 2023). While a drainage network model incorporating every real-world element may require significant computational resources, it does not automatically guarantee greater accuracy (Dobson et al., 2022; Krebs et al., 2014). To save computational power, the majority of prior research focuses on modeling key junctions and often omits detailed real-world inlet components like grates and curb openings in their stylized drainage network model (Dong et al., 2023; Leitão et al., 2010). Therefore, if input data for the actual urban drainage system is unavailable, an alternative is to construct a stylized drainage network model based on the local sewer design standards and manuals (for detailed methods, refer to Section 3.2.).

3.2. Hydrological model construction

In order to construct urban pluvial flood modeling, we need to utilize both the stormwater model and the overland flow model. For the sake of model simplicity, we suggest coupling a one-dimensional stormwater model and a rapid flood spreading model (1D-RFSM) for this framework. For the stormwater model part, essential inputs include rainfall data, terrain data, LULC, hydrologic soil group, and the urban drainage network. The output of the stormwater model consists of flood volume, flood duration, flooding rate, and flooded nodes, which are then used as inputs for the overland flow model part.

First, it is essential to construct a model of the drainage network. If geographical data for the drainage system in the target area is already available, it can be used directly. Alternatively, for a minimum viable model, users can consolidate local or tributary sewers with the same flow direction to build a stylized drainage network model that emphasizes the trunk or intercepting sewers (Davidsen et al., 2017; Leitão et al., 2010). If geographical data for the drainage system in the target area is unavailable, a stylized drainage network model can be constructed based on the principles outlined in the local sewer design manual. As a good rule of thumb, junctions were positioned at every node and at points where the conduits curved. In straight sections, the spacing between junctions should not exceed the maximum interval distance, defined by the sewer design standards of the local government. Since most urban drainage systems transport sewage by gravity, it is necessary to know the slope of the conduit, specifically the depth (invert elevation) of its starting and ending junctions, to calculate storm drain flow (Schmitt et al., 2004). However, if the invert elevation data for each junction is unknown, the drainage network model can be completed by applying the pipe design slope according to the conduit diameter, as provided in the sewer design standards of the local government.

Secondly, to estimate surface runoff from rainfall data, the initial step is to delineate subcatchments according to the drainage network map (if available) within the target area. However, if a drainage network map is not available, subcatchments can be delineated based on the grid zone or road network in urban areas, since public sewers are often installed under the road layout (Amaguchi et al., 2012). Using GIS, the surface slope and imperviousness of each subcatchments can be derived based on terrain data. Subsequently, the NRCS-CN method is employed to estimate rainfall-induced runoff or infiltration, commonly used in ungauged watersheds. Although it lacks a connection to real-world hydrological processes compared to physically distributed models, its use of a few parameters allows for faster and sometimes even more accurate results (Sitterson et al., 2017). Using Table S.1, the NRCS-CN values can be derived through LULC as well as hydrologic soil groups.

Third, to calculate the combined sewer inflow into the target area, identify the junctions where external inflows come from the corresponding adjacent subcatchments. External inflow from adjacent subcatchments refers to the base sanitary flow (dry weather inflow) on clear days, and during rainfall, it means the sum of that base sanitary flow and the rainfall-induced surface runoff (wet weather inflow). For the base sanitary flow coming from neighboring subcatchments, the estimation was performed by taking the product of the conduit's cross-sectional area (m^2), the average sewage carrying capacity (%), and the flow velocity (m/s). If the flow velocity cannot be obtained, the design flow velocity can be used instead. The rainfall-induced surface runoff from each adjacent subcatchment is estimated using the modified rational method, which is widely used for rapid flood assessment and determining the design capacity of urban drainage systems (Aron & Kibler, 1990). It is posited that the surface runoff from the outlet of each adjacent subcatchment all flows into the target area's urban drainage network.

Fourthly, run a 1D stormwater model simulation. The governing equation for routing both sewer flow and unsteady surface flow is the Saint-Venant equation, solved through the application of the finite difference method (Schmitt et al., 2004). As input, a rainfall hyetograph is used, and intermediate results for surcharged overflow from each node are generated, such as total flood volume, flood duration, and peak flooding rate and time.

For the overland flow modeling, we suggest using the simplest flood modeling approach, the Rapid Flood Spreading Model (RFSM), as part of this framework. RFSM only takes the DSM and the surcharged overflow volume derived from the stormwater model as its inputs, and yields final outputs such as flood inundation extent and depth (Bernini & Franchini, 2013). Despite its limitations, such as not considering overland flow velocity and being only capable of ascertaining the boundaries of the flood inundation area, RFSM shows acceptable accuracy for simulating an individual rainfall event compared to more complex flood modeling approaches, like 1D or 2D models (Jamali et al., 2019). Therefore, RFSM is commonly used in areas where rapid flood assessment is required for real-time forecasting, early warning, and emergency operations, as well as in regions where the available input data is not enough for conducting higher-dimensional flood models (Bulti & Abebe, 2020). The procedure of flood spreading begins by filling local depressions and then proceeds to overflow floodwater into adjacent grid cells until no more floodwater is left (Jamali et al., 2018). We can determine the surcharge-induced inundation depth by adding up the flood volume on the DSM of the target area. By repeating this process for multiple flood events, the relationship between total flood volume and flood inundation depth can be derived through nonlinear regression analysis, resulting in a pre-defined flood volume-depth curve. Using this curve can dramatically reduce computational times to a matter of seconds, enabling immediate near real-time responses. For specific parameter values and model settings, please refer to Section 4.2.

3.3. Model verification

As previously discussed, due to the lack of observed data on the dynamic change of urban pluvial flooding extent or depth, we conduct model calibration and validation using multifaceted data and methods. To verify the stormwater model, data about the sewer depth or flow rate in the conduit is required. This is possible only if the sewer management entity is conducting regular monitoring. However, the overland flow model can be validated through various means such as historical flood extents, flood scene images, flood marks, or anecdotal evidence. Photographs of in situ flood situations can often be obtained through social media, while historical flood extents and flood marks are usually provided by local governments or accredited surveyors (Le Coz et al., 2016). Anecdotal evidence, such as how high the water rose or which roads were inundated, can be gathered through interviews with residents or posts on social media (Wang et al., 2018).

The calibration process is conducted for the parameter sets related to the characteristics of the urban drainage system, within the ranges confined by previous studies outlined in Table 1. The optimum parameter values were calibrated in small increments (i.e., 0.001 unit) to minimize the error between the observed values and the model output. The goodness-of-fit measure for calibration and validation varies depending on the target variable and observed data.

Firstly, one method of calibration and validation involves using historical flood extents, which are geographic data represented by polygon features. In order to verify the geometric similarity between two polygons, the Lee-Sallee index is often used in hydrology and flood model validation (P.D Bates & De Roo, 2000; Bates et al., 2005; Bojórquez-Tapia et al., 2009; Horritt & Bates, 2002; Smith et al., 2012). The Lee-Sallee index, formulated by Lee and Sallee in 1970, uses Eq. 1 to assess the spatial overlap between observed and modeled flood inundation areas, designated as either A or B (Lee & Sallee, 1970). This explicit measure provides a value that ranges between 0 and 1, with closer to 1 meaning that A and B completely coincide (Clarke & Gaydos, 1998; Jantz & Goetz, 2005).

$$Lee-Sallee\ index = \frac{A \cap B}{A \cup B} \tag{1}$$

Secondly, another method is to use flood scene images for visual comparison. It is able to perform a plausibility check by matching the key features of the actual flood inundation extent and depth with the simulated flood map, utilizing geo-tagged pictures from social media (Brouwer et al., 2017; Fohringer et al., 2015).

Thirdly, one can use flood marks or anecdotal evidence, which often have specific values. Flood marks indicate the maximum inundation depth, and anecdotal evidence related to flood damage usually claims fixed values rather than changes over time (Rollason et al., 2018).

Additionally, due to the scarcity of observed flood-related data, another option is to validate the stormwater model using sewer level monitoring data in the pipes, rather than directly verifying the final output from the overland flow model. In such cases, popular goodness-of-fit measures used in hydrology, such as the R-squared, Root Mean Square Error (RMSE), Nash–Sutcliffe Efficiency (NSE), the RMSE-observations standard deviation ratio (RSR), and Percent Bias (PBIAS), can be applied (Moriasi et al., 2007). Among them, we adopted the NSE and the RSR in this study.

First of all, the NSE serves as a standardized indicator that measures model accuracy by contrasting the variance of the model's residuals against the variance of the observations (Nash & Sutcliffe, 1970). NSE is widely accepted as a goodness-of-fit measure in hydrological model simulations because it is less sensitive to outliers compared to correlation-based measures (e.g., R-squared) (Moriasi et al., 2007; Shadmehri Toosi et al., 2020). The mathematical formula of NSE is presented in Eq. 2:

$$NSE = 1 - \frac{\sum_{i=1}^{n}(O_i - P_i)^2}{\sum_{i=1}^{n}(O_i - \overline{O})^2} \tag{2}$$

where O_i indicates the observed value at the ith position, P_i is the predicted value at the ith position, $\overline{O}$ represents the mean of the observed values, n is the total number of observations.

Next, among the model evaluation statistics concerning the error index, the RMSE has been commonly used to estimate residual variance. However, to address the drawbacks of RMSE's unnormalized values and case-specific units, Moriasi et al. (2007) developed a standardized and dimensionless index called the RMSE-observations standard deviation ratio (RSR). The RSR is literally defined as the ratio of the RMSE to the standard deviation of observed data (Yaqub et al., 2020). RSR is expressed by the following equation (Eq. 3):

$$RSR = \frac{RMSE}{STDEV_{obs}} = \frac{\sqrt{\sum_{i=1}^{n}(O_i - P_i)^2}}{\sqrt{\sum_{i=1}^{n}(O_i - \overline{O})^2}} \tag{3}$$

Table 1
Subcatchment and conduit properties and parameter values in SWMM.

Parameter	Description	Surface (Conduit material) type	Range of typical values	Reference
Subcatchments				
Roughness	Manning's n for overland flow on (im)pervious subcatchment	Smooth asphalt	0.011	(McCuen et al., 1996)
		Smooth concrete	0.012	
		Fallow soils	0.05	
		Cultivated soils	[0.06, 0.17]	
		Range	0.13	
		Grass	[0.15, 0.41]	
Dstore (mm)	Depression storage depth on (im) pervious subcatchment	Impervious surfaces	[1.27, 2.54]	(ASCE, 1992)
		Lawns	[2.54, 5.08]	
		Pasture	5.08	
		Forest litter	7.62	
C	Runoff coefficient of the Rational Formula	Downtown area	[0.7, 0.95]	(TxDOT, 1994)
		Residential area	[0.3, 0.75]	
		Suburban area	[0.35, 0.4]	
		Industrial area	[0.3, 0.9]	
		Park	[0.1, 0.25]	
		Lawns	[0.05, 0.35]	
		Streets	[0.7, 0.95]	
Conduits				
Roughness	Manning's n for closed conduit	Cast iron pipe	[0.011, 0.015]	(ASCE, 1982)
		Concrete pipe	[0.011, 0.015]	
		Corrugated-metal pipe	[0.011, 0.026]	
		Plastic pipe	[0.011, 0.015]	
		Vitrified clay	[0.011, 0.017]	

3.4. Model experiment

3.4.1. Green infrastructure strategy

Users can explore the most effective pluvial flood mitigation and climate change adaptation strategies by conducting various types of simulation experiments with a constructed urban pluvial flood model. Firstly, there is a strategy that uses green infrastructure, also known as

low impact development (LID). Users can employ mixed LID practices to reduce pluvial flood volume by enhancing stormwater retention and detention (Tansar et al., 2022). For example, SWMM version 5.1 and above offers 8 types of LID control modules for different surface runoff routing (bio-retention, rain gardens, green roofs, infiltration trenches, permeable pavement, rain barrels, rooftop disconnection, vegetative swales). Through the installation of LID, one can improve urban stormwater management by diverting the temporal and spatial runoff distribution.

3.4.2. Grey infrastructure strategy

Secondly, another approach involves retrofitting traditional grey infrastructure. This approach focuses on engineering solutions like enhancing the carrying capacity of the existing drainage system, such as expanding the diameter of pipes and increasing the pipe slope. Additionally, new conduits can be added for better stormwater management. A notable example would be the deep tunnel sewer systems that are being adopted in many mega-cities worldwide (Chen et al., 2021; Park et al., 2024). These systems serve as an advanced engineering solution for managing large volumes of stormwater and wastewater. By exploring various network configurations of urban drainage system, users can identify the optimal measures to alleviate the bottleneck and minimize the surcharged overflow volume.

3.4.3. Collective action strategy

The third strategy focuses on enhancing a community's flood resilience through collective action, such as the voluntary organization and cooperation among citizens (McEwen et al., 2018; Mees et al., 2016). This can involve installing water barriers or temporary sandbags, as well as cleaning the inlets of the urban drainage system to ensure they remain unclogged. Or, sharing information about flooded roads to control traffic around dangerous areas and to identify detour routes can help vulnerable road users evacuate safely. On a personal level, individuals can be motivated by immediately seeing how and to what extent their efforts contribute to reducing flood damage. On the community level, scientific analysis can determine the number of participants needed for effective stormwater management, thereby aiding in the organization and mobilization of resources.

3.4.4. Climate change scenarios

Lastly, users can explore and compare above stormwater management strategies by applying various climate change scenarios to predict the impact of uncertain heavy rains in future. Additionally, by testing different return periods of design rainfall under climate change scenarios, they can identify the tipping points for each urban pluvial flood mitigation strategy. The method for determining design rainfall entails aligning historical precipitation data with a probability distribution function, followed by generating an Intensity-Duration-Frequency (IDF) curve (Esteves, 2013). We estimated the volume of rainfall expected for the selected return period, taking into account various climate change models and scenarios. To generate hydrographs based on that interest design rainfall, we opted to use the alternating block method within this framework (Gersonius et al., 2012).

4. Case study

In this section, we selected the Gangnam region of Seoul, South Korea, as a case study to showcase the participatory flood modeling framework. We used popular open source software and provided detailed guidelines on the essential techniques step-by-step, with the exception for real human participant recruitment from the study area. The Gangnam region is one of the largest central business districts and the most expensive residential areas in South Korea. In the Gangnam region, we developed a urban pluvial flood model for Seocho 2-dong, Seocho-gu. This case study site covers an area of 1.02 km^2 and has experienced pluvial flooding biennially for the last decade. In 2022, the study area experienced a 100-year heavy rainfall with a maximum intensity of 92 mm/h, leading to the most severe flood damage ever recorded in Seoul, South Korea. The target area has a U-shaped topography that is lower than its surroundings. Therefore, the runoff from adjacent regions initially gathers in the target area before being discharged into the Han River via a northwestern tributary (Fig. 4). However, the recurring disasters are not solely due to the vulnerable conditions of topography, but also to the insufficient capacity and ill-designed drainage systems of the target area (Park et al., 2024). In this case study, we used EPA SWMM 5.2 for the rainfall-runoff and sewer flow simulation, and QGIS 3.28.10 for the rapid flood spreading simulation; both are leading open source software.

4.1. Input data preparation

According to the framework, we describe how we acquired each of the following input data listed in Table 2.

For rainfall data (Fig. 4.D), we used hourly precipitation records obtained from the Korea Meteorological Administration website (https://data.kma.go.kr). The rain gauge is located at latitude 37.48462°N, longitude 127.02601°E, within the target area. For the terrain data, we downloaded a 1:1000 digital topographic map from the National Geographic Information Institute of Korea website (http://map.ngii.go.kr) and converted it into a 5m grid-size Digital Terrain Model (DTM) using the Triangulated Irregular Network (TIN) interpolation method (Fig. 4.A). We created the Digital Surface Model (DSM) using spatial information such as building area and height (Fig. 4.B), which was obtained from Korea's National Spatial Data Infrastructure portal (https://www.vworld.kr). For the detailed land cover map (Fig. 4.C), we utilized the data from the Environmental Geographic Information Service website of the Ministry of Environment of Korea (https://egis.me.go.kr). Using the slope and zonal statistics plugins in QGIS, we calculated the average surface slope for each subcatchment from the DTM. Based on the land cover map, we assessed the level of imperviousness for each subcatchment; on average, the fraction of the impermeable surface area is 79.8% (Table S.2). The hydrologic soil groups for the target area are classified as Group D (Bae et al., 2003).

If you have geographic data (for example, in *.shp file format) about the urban drainage system, it is convenient to use the 'Generate SWMM inp' plugin in QGIS to convert it into *.inp file, which is the input file format for SWMM (Schilling & Tränckner, 2022). For the study area, the layout of the trunk sewer is publicly available only in printed format for security reasons, as depicted in Fig. 4.A (Seoul Metropolitan Government, 2018). In this study, we constructed a stylized drainage network model based on the trunk sewer layout, supplemented with the sewer design standards published by the Ministry of Environment of Korea (Refer to Section 4.2).

Finally, we need orthophotos of the study area to render the digital twin model for virtual reality, which can also be downloaded from the National Geographic Information Institute of Korea website.

4.2. Hydrological model construction

We provide a detailed guideline of the coupling process between the 1D stormwater model and the rapid flood spreading model (RFSM) based on the framework.

To construct the stormwater model part, we first constructed a stylized drainage network model based on Korea's sewer design standards (Ministry of Environment, 2022) as illustrated in Fig. 5.A. Given the grid zoning in the Gangnam region, we position junctions in SWMM where conduits intersect or change direction, and we also ensure that a junction is placed at least every 200m along straight sections, following the sewer design standards. We assumed the conduit's shape to be a closed rectangular box (identical to reality), and set its cross-sectional dimensions to the average found in Seocho District, which are 2.2m (W) * 2.2m (H) (Fig. 5.C). The maximum soil depth (from the crown to

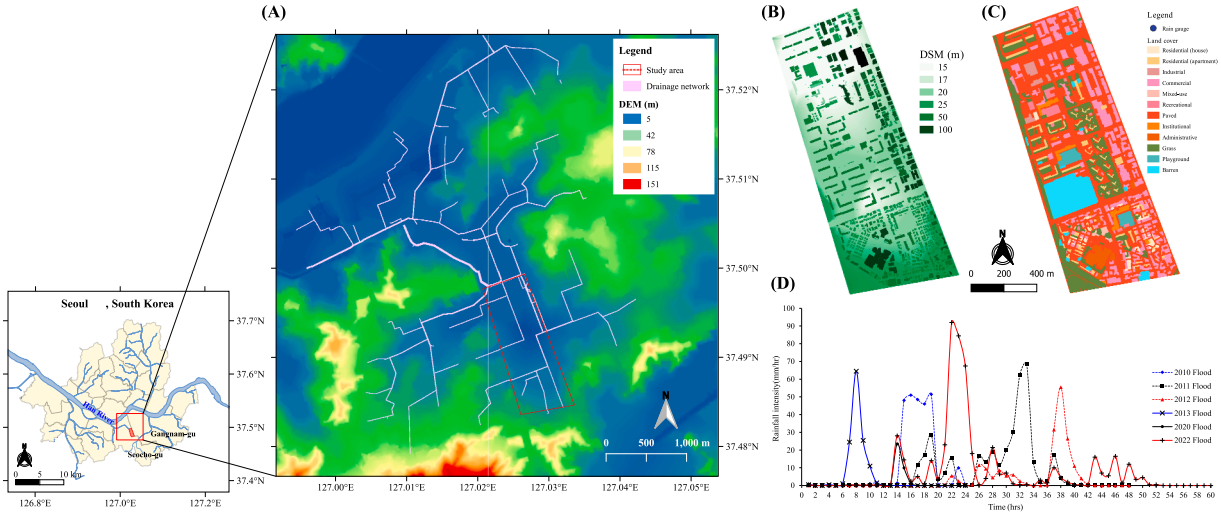


Fig. 4. Input data used for constructing an urban pluvial flood model for Seocho 2-dong, Seoul, South Korea: (A) digital terrain model and urban drainage network, (B) digital surface model, (C) land cover map, and (D) hyetographs that caused historical floods in the target area.

Table 2
Input data summary of its types and sources used in this study.

Input data	Description	Data type	Source
Input data preparation			
Rainfall data	Hourly precipitation (mm)	Numerical	Korea Meteorological Administration
Terrain data	Digital topographic map (1:1000)	Geographical	National Geographic Information Institute of Korea
Building data	GIS building data	Geographical	National Spatial Data Infrastructure of Korea
Land use/land cover	Detailed land cover map	Geographical	Environmental Geographic Information Service of Korea
Soil data	Hydrologic soil group	Categorical	(Bae et al., 2003)
Urban drainage network data	Trunk sewer layout	Geographical	(Seoul Metropolitan Government, 2018)
	Sewer design standards	Text	(Ministry of Environment, 2022)
Orthophoto	Orthophoto (25cm)	Geographical	National Geographic Information Institute of Korea
Model verification			
Historical flood extent	Peak flood inundation extent	Geographical	Seoul City public database
Sewer level data	Continuously monitored sewer level every minute (m)	Numerical	Seoul City public database
Flood scene images	Volunteered geographic information	Photograph	Social media

the ground) was established at 1m, corresponding to the average in Seocho District. The pipe slope was set at 1.4 ‰, in accordance with the Sewerage Improvement Master Plan of Seoul City, specifically for a pipe design slope with an internal diameter of 2.2m (Seoul Metropolitan Government, 2018). The roughness of the conduit, represented by Manning's n for closed conduit, was calibrated within the parameter value range suggested in prior research based on the conduit material type (refer to Section 4.3).

Secondly, to estimate the surface runoff within the target area, we delineated the subcatchment based on the constructed drainage network model. In SWMM, we identified outlets for each subcatchment and calculated the overland flow paths to incorporate them (Table S.2). The infiltration method was set to NRCS-CN, the drying time was designated to span 7 days, and we assumed that the percentage of impervious area with no depression storage (%Zero-Imperv) was zero. Using Table S.1, we assigned CN values for each land cover type with respect to soil properties, and calculated the weighted average CN for each subcatchment. We input the derived values into SWMM as attributes for each subcatchment. Additional subcatchment properties and parameter values were set according to Table S.2 or determined through a calibration process to achieve optimal values (refer to Section 4.3).

Thirdly, as shown in Fig. 5.B, we identified the junctions within the drainage network model where external inflow occurs and delineated the adjacent subcatchments. We applied the modified rational method to estimate runoff from adjacent subcatchments and fine-tuned the runoff coefficient in the rational formula. In Seoul, the rational formula for urban runoff is restricted to areas under 5 km^2 (Seoul Metropolitan Government, 2009); our study area complies with this condition. The base sanitary flow from each adjacent subcatchment was calculated using the conduit's cross-sectional area, the rainy season average conduit capacity of 18.1% from 2018-2022, and the design velocity in Table S.3.

Fourthly, to calculate the surcharged hydrographs at each junction, we simulated the 1D stormwater model using SWMM (Fig. 6). We used the Dynamic Wave model for flow routing with the Extran surcharge method under the assumption of no ponding areas at the junctions. The simulation was executed with a routing time-step of 1 second and a reporting time-step of 1 minute. For the overland flow model, we used the flood hydrograph derived above and the DSM to conduct the RFSM. To do so, the depth of flood inundation was calculated by filling the surcharge volume on the target area's DSM over time using the raster surface volume tool in QGIS. By iterating this method, a correlation between the total volume of floodwater and the depth of flood inundation can be derived, as well as a correlation between the inundation area and the mean flood depth, as illustrated in Fig. 7. These pre-defined relationships can significantly save computation time in model experiments conducted concurrently by multiple users in online environments with limited computing resources.

4.3. Model verification

Given the poor quantity and quality of observed pluvial flood-related data, we adopted a double validation approach by using two or more available datasets for cross-checking: geometric comparison through flood inundation extents, quantitative comparison with monitored

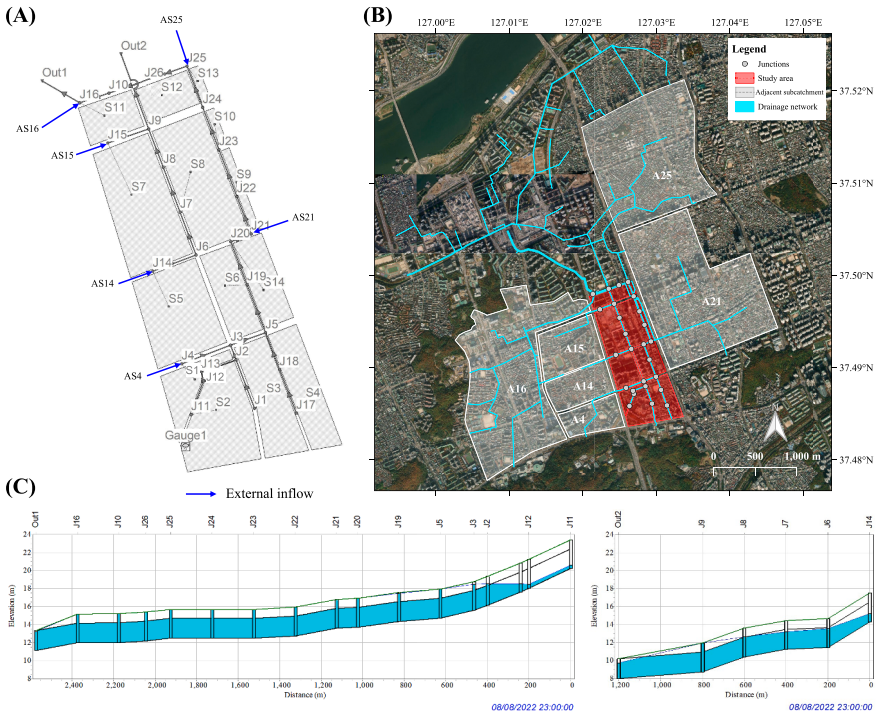


Fig. 5. Constructed drainage network model of the target area: (A) stylized drainage network model representing subcatchment, junctions, and inflow nodes, (B) adjacent subcatchments drain stormwater to the target area, and (C) longitudinal profiles of two separate trunk pipelines discharging to Out1 and Out2.

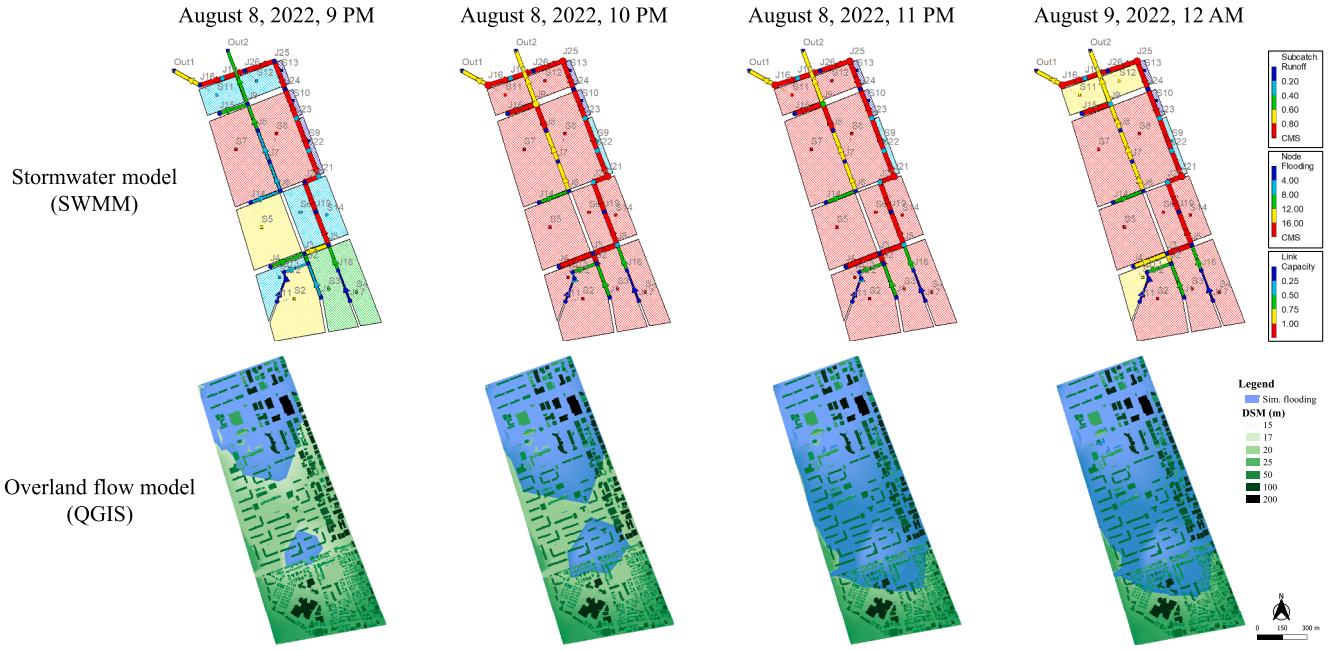


Fig. 6. Dynamic simulation results of stormwater model and overland flow model respectively over time for the 2022 flood in the target area: The stormwater model returns the results for surface runoff, surcharged nodes, and the capacity of the drainage conduit. The overland flow model indicates changes in the inundated area.

sewer level data, and visual comparison using flood scene images (Table 2). For the flood data used in model verification, we selected rainfall events that caused significant flood damage in the target area, provided that at least two different types of datasets were available. To ensure reliability, we conducted double validation for the flood events of 2010 and 2022 in the target area, respectively. Depending on the types of available data, dual validation combining geometric and visual comparisons was conducted for the 2010 flood, while quantitative and visual comparisons were performed together for the 2022 flood.

First, for the double validation of the 2010 flood event, we used the 2010 flood extent map from the Seoul City public database (https://data.seoul.go.kr) as the primary data source for calibration with the output from the overland flow model (Fig. 8). We calibrated the parameter values listed in Table S.4 to minimize the geographical difference between observed and simulated flood inundation extents. In other words, we aimed to maximize the Lee-Sallee index between those two polygons using Eq. 1, as depicted in Fig. 8.A. There is no standard for the performance rating of the Lee-Sallee index that has been

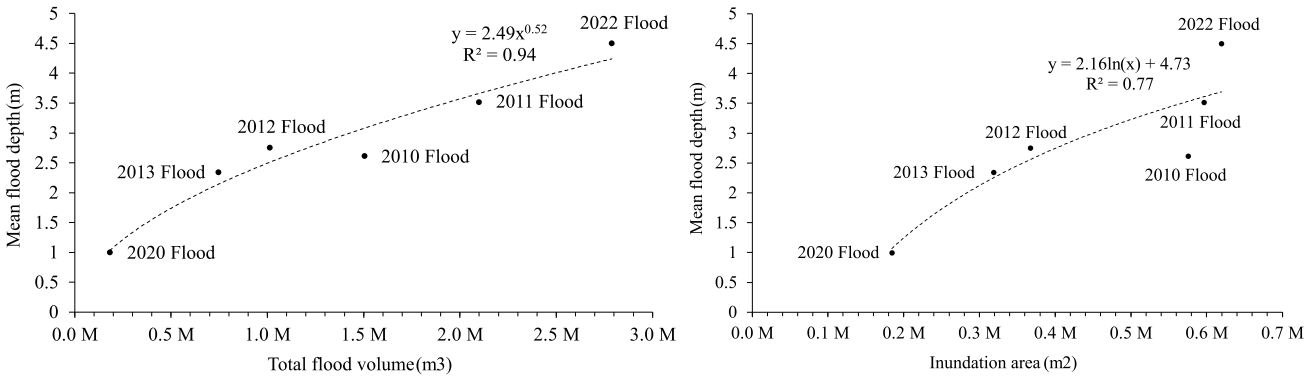


Fig. 7. Total flood volume–mean flood depth curve (left) and inundation area–mean flood depth curve (right) for Seocho 2-dong, Seoul, South Korea.

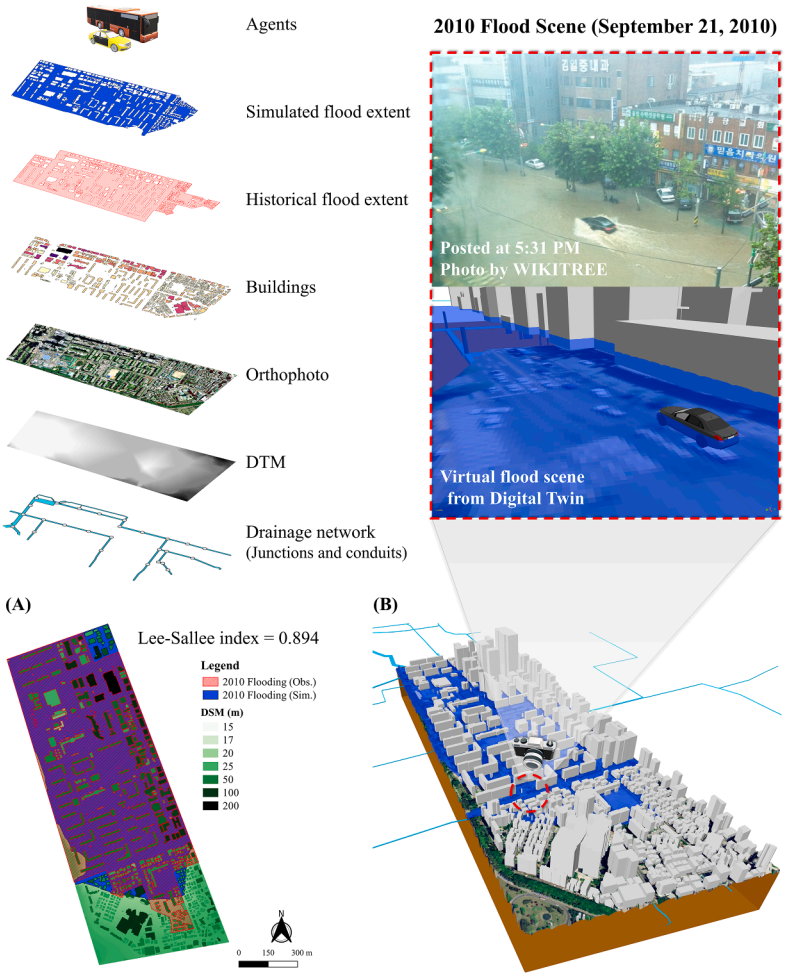


Fig. 8. Double validation of the urban flood model using the 2010 flooding event in the study area included: (A) Geometric comparison of the observed and simulated flood inundation extents; and (B) Visual comparison of the actual flood scene (Photo by WIKITREE) with the virtual flood scene from the digital twin model.

generally agreed upon yet, but traditionally, a match of 30% or more has been considered good in comparisons between binary layers (Clarke & Gaydos, 1998). A recent study proposed the following performance benchmarks for the Lee-Sallee index: "Excellent" for values above 0.4, "Good" for values above 0.3, "Fair" for values above 0.2, and "Poor" for values above 0.1 (et al., 2019). The final Lee-Sallee index value of our model shows 89.4% accuracy, which can be considered a good performance when compared to the results from other key papers using this method for peak inundation validation: 77.3-81.6% (Bates & De Roo, 2000), 37.4-65.3% (Horritt & Bates, 2002), and 54-91% (Bates et al., 2005).

Next, as a second data source, we conducted a visual comparison using volunteered geographic information (VGI) in the form of flood scene images taken by a citizen during the 2010 flood in the target area and shared online. Fig. 8.B shows a comparison of an actual flood scene image and a virtual flood scene generated by the digital twin model. Digital twin models can incorporate agents, such as humans or cars, as reference points for comparing flood inundation depths and extents (Yan

et al., 2024). Additionally, users can navigate the virtual flood scene from various angles and heights, proving its effectiveness in the visual comparison of participatory flood models. However, social media was in its infancy during the 2010 event, resulting in a scarcity of VGI with limited geo-locations. Since then, the explosion of social media use has enabled more extensive visual comparison at numerous points, thus becoming one of the reliable methods of flood model verification.

Secondly, for the double validation of the 2022 flood event, we performed a formal goodness-of-fit test to quantitatively compare the observed sewer level data with the simulated sewer depth, an intermediate output of the stormwater model before coupling with the overland flow model. The monitored sewer level data for the study area has been available every minute in a few specific conduits since 2012 from the Seoul City public database (https://data.seoul.go.kr). Among them, we used the observed values from a conduit near one of the study area's outlets (J9-Out2), as shown in Fig. 9, and conducted quantitative validation with the 2022 flood rainfall, which resulted in the most significant flood damage ever recorded. We evaluated two measures of goodness-of-fit, specifically NSE and RSR, broadly utilized in the field of hydrology.

The NSE, a measure of model accuracy compared to the observed data, ranges from negative infinity to 1. The higher the NSE value (closer to 1), the better it captures the actual behavior observed in the real world. Moriasi and colleagues (2007) (D. N. Moriasi et al., 2007) set forth general performance benchmarks for hydrological models with monthly time steps, categorizing an NSE greater than 0.75 as "Very Good", those between 0.65 and 0.75 as "Good", and those above 0.50 as "Satisfactory". However, our model operates on a minute-by-minute basis, implying that the performance criteria designed for monthly intervals might be overly strict, given that model accuracy generally diminishes with shorter time steps (Engel et al., 2007). As far as we are aware, there are no established performance benchmarks for models operating on a minute-by-minute basis, prompting us to follow the standards for monthly models. Consequently, we have cautiously assumed that our model's performance is on par with, or superior to, that of monthly models for the same index value. Using Eq. 2, the NSE value of the stormwater model for the minute-based sewer level data is evaluated as 0.52, which is considered satisfactory or better.

The RSR, a normalized indicator of error variance, varies from 0 to positive infinity. The closer the RSR value is to 0, the fewer errors there are in the model simulation. According to Moriasi et al. (2007), the suggested performance benchmarks for the RSR with monthly time steps classify an RSR less than 0.5 as "Very Good", those between 0.5 and 0.6 as "Good", and those between 0.6 and 0.7 as "Satisfactory". Again, we have conservatively adopted the criteria for monthly time steps to assess the performance of our minute-by-minute model. Using Eq. 3, the derived RSR value is 0.69, indicating that the model performance is at least satisfactory in terms of the residual variation of sewer level data. In summary, both goodness-of-fit measures demonstrate the performance of the stormwater model as above satisfactory.

As a second validation method, we also conducted a visual comparison between VGI and the digital twin model. We used actual flood scene images taken at four different locations in the study area during the 2022 flood for plausibility checks through visual comparison. Assuming that the time the photos were first posted on social media is the time when the flood scene was captured, this appears to be a situation just before the peak flood. It was compared with the simulated flood inundation map for 10 p.m. on August 8, 2022. Although it is difficult to make a strict comparison due to the possible difference between the actual record time and the upload time, it can be said that the pattern of inundation extent and depth was well captured at each point. Technically, we utilize the Qgis2threejs plugin (version 2.7.1) in QGIS to construct a 3D flood inundation map. This can be exported in the form of a webpage written in HTML and JavaScript, creating an interactive 3D platform that can be uploaded online, as shown in Fig. 9.

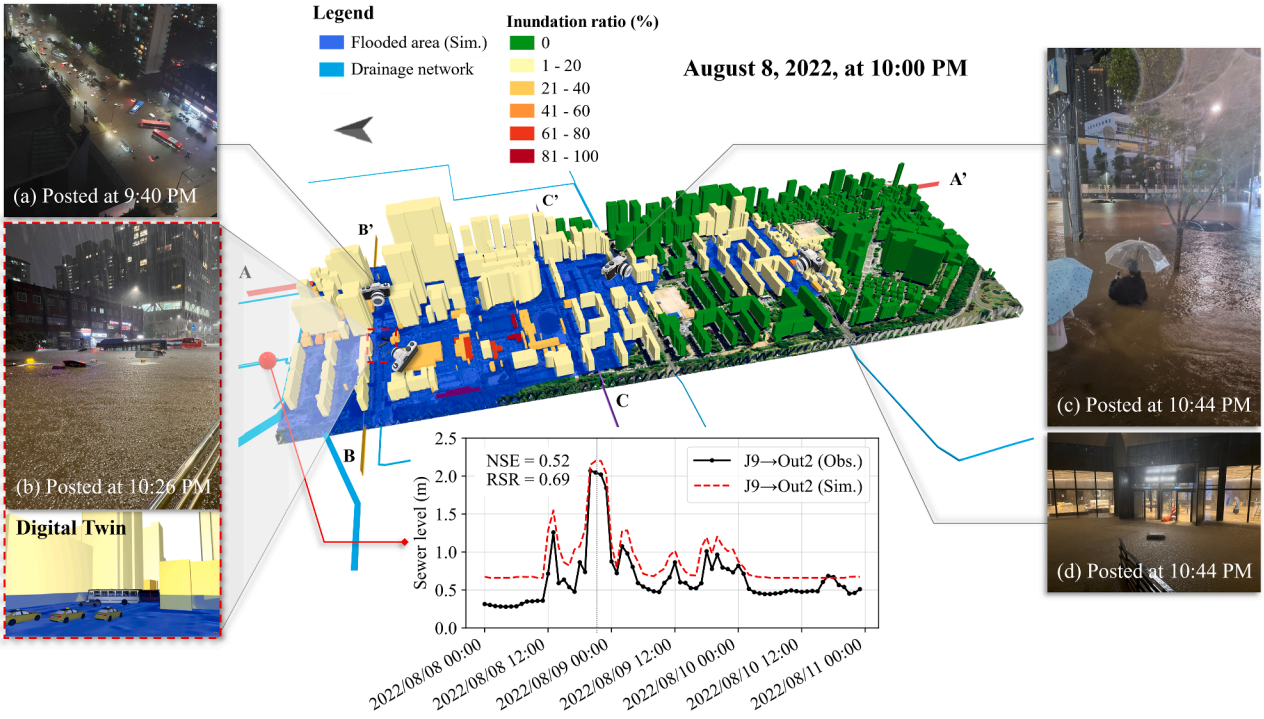


Fig. 9. Double validation of the urban flood model using the 2022 flooding event in the target area involved: (1) evaluating goodness-of-fit measures (i.e., NSE, RSR) of the stormwater model by comparing observed and simulated sewer levels (bottom-middle); and (2) conducting a visual comparison between the simulated flood inundation map in three dimensions and actual geolocated images shared on social media around 10 p.m. on August 8, 2022 (Photos by (a) Alfredo (@GetBackMyDream), (b) TTC (@TTC_Trader), and (c and d) The Korea Herald (@TheKoreaHerald) / Twitter). Navigate the digital twin model through your web browser on desktop or mobile devices using the following link: http://sampark.kr/floodmap/gangnam/f2022/10pm/

4.4. Model experiment

We demonstrate experiments for green infrastructure, grey infrastructure, and collective action strategies using the constructed urban pluvial flood model, based on extreme rainfall events driven by climate change scenarios.

4.4.1. Green infrastructure strategy

Using the 2010 flood rainfall as a case study, we applied three types of LID practice scenarios—namely, permeable pavement, bio-retention, and green roofs—to 50% of the area of impervious surfaces, green spaces, and buildings, respectively. The parameter values for LID implemented in the SWMM model can be found in Table S.5. Fig. 10 displays the dynamic retention and detention effects of each LID practice in runoff reduction. Fig. 10.A illustrates that the majority of direct runoff from or in-situ rainfall on LID practices could have been effectively retained through the use of combined permeable pavement and bio-retention. Additionally, Fig. 10.B shows that the water-holding capacity of green roofs varied over time in two distinct sections: first, in the surface and soil layers where ponded water was fully drained within 3 weeks, and second, in the drainage mat, where the detained water was progressively released over a span of 40 days. In terms of the total volume of retained or detained stormwater, permeable pavement contributed to 74% and bio-retention made up 15% for the retention volume, as indicated by the total area under the infiltration loss rate curve for each LID practice in Fig. 10.A. The detention volume from green roofs accounted for 11%, as shown by the sum of the maximum ponded and detained water volume in Fig. 10.B. In conclusion, LIDs achieved an 8.4% reduction in total flood volume of the 2010 flood event compared to the baseline without the application of LID practices.

4.4.2. Grey infrastructure strategy

As a strategy to improve grey infrastructure, we devised scenarios that double the conduit capacity in sections of the existing drainage network that frequently experience bottlenecks. We then compared their effectiveness in reducing flood risks using the rainfall data from the 2010 flood (Table 3). Comparing the Grey1 and Grey3 scenarios, Grey1 reduced the flood volume by 35.8% by enlarging the conduit capacity of only two links, whereas Grey3, despite improving a longer section of four links, had a negligible effect on reducing flood volume (-0.4%). Accordingly, it can be seen that focusing on critical bottleneck sections, regardless of the overall length upgraded, is the most effective strategy for mitigating urban flood risks. In this case study, the J10-Out1 section serves as the leverage point. Likewise, the Grey2 scenario, which includes the J10-Out1 section and extends upstream to encompass a total of four sections for diameter enlargement, is analyzed to potentially reduce flood volume by up to 40.3% compared to the baseline. In summary, if efficiency in terms of flood reduction per link is considered, the Grey1 scenario would be the optimal strategy, and if the goal is to minimize the total flood volume, the Grey2 scenario would be the better option. Based on the results of scenario analysis, the public can engage in discussions with various stakeholders about policies for retrofitting the most effective sections of the drainage network.

4.4.3. Collective action strategy

Through participatory urban flood modeling, citizens can find areas where they can voluntarily contribute to flood risk reduction. For instance, as discussed in Section 3.4.3, one way to avoid flood damage in the community is to install water barriers or temporary sandbags collectively. Participatory modeling can help them to identify the pluvial flood-prone area and analyze the required length and height of water barriers or sandbags. We conducted a case study on collective action strategies utilizing the 2022 flood event, which featured a building that actually avoided flooding with water barriers.

The flood inundation simulation results for the 2022 flood rainfall around peak flood period (August 8, 2022 at 11 p.m.) are represented in Fig. 11. The right-hand picture in Fig. 11 depicts the building (marked with a star) that successfully prevented flood damage in 2022 through the voluntary installation of a flood barrier. Based on the simulation results, one can calculate the required dimensions of water barriers to protect buildings from urban pluvial flooding by estimating the inundation depth. The flood depth around this building at that time was simulated to be 1.38m (refer to C-C' cross-section). The average inundation percentage across all buildings is calculated to be 4.9% according to Eq. 4, while it stands at 17.3% when considering only the buildings that were flooded.

$$\textit{Inundation ratio}\ (\%) = \frac{\textit{Flood depth}}{\textit{Building height}} * 100 \tag{4}$$

Such information can assist in determining the height of the flood barrier and guide the voluntary installation of flood barriers or sandbagging in response to the anticipated risk of urban flooding. Through this approach, citizens can voluntarily mobilize the community's resources and prioritize collective actions, such as sending relief goods and volunteer rescue teams to the buildings most at risk of flood damage. Eventually, participatory modeling that supports collective action is a process that facilitates the adaptive capacity of communities, thereby enhancing community disaster resilience.

4.4.4. Climate change scenario

For the climate change model, we employed the RCP4.5 and RCP8.5 scenarios provided by the Global/Regional Integrated Model System

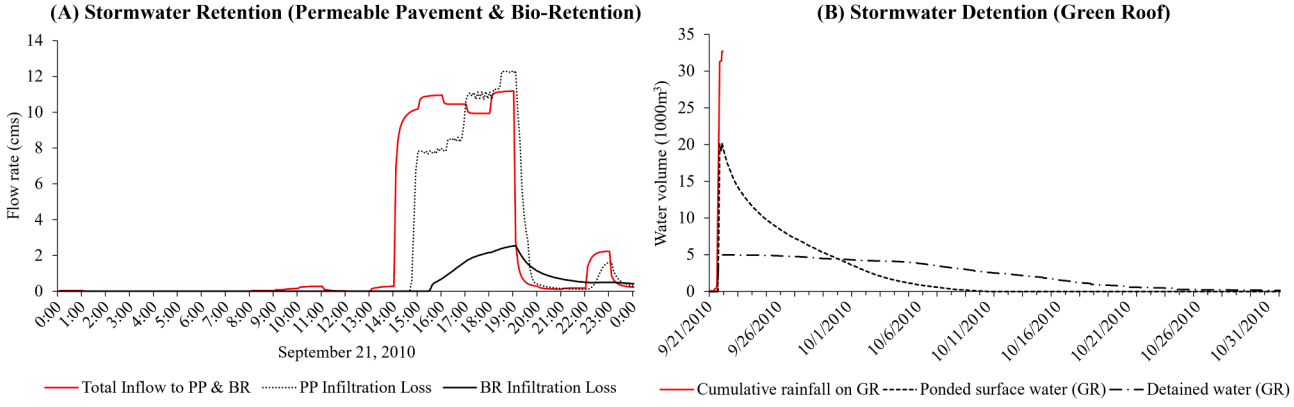
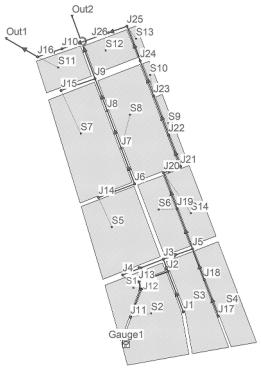
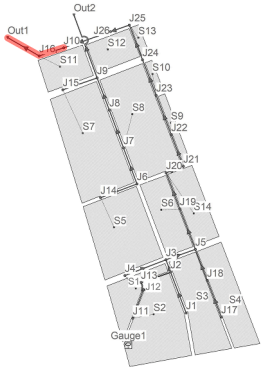
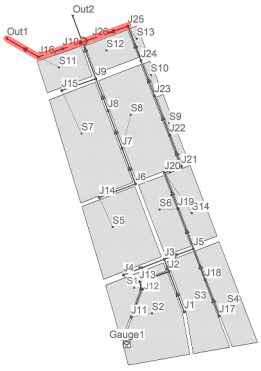
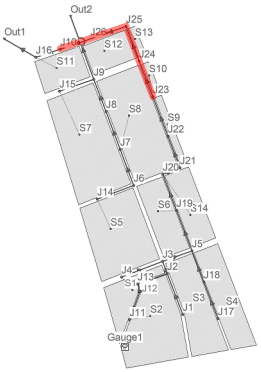


Fig. 10. Performance of LID practices in retaining and detaining runoff over time: (A) For the retention effect, examine the inflow rate and infiltration loss rate of permeable pavement (PP) and bio-retention (BR) on September 21, 2010. (B) To assess detention effect, track ponded and detained water volume changes in green roofs (GR) for 40 days post-rainfall.

Table 3
Scenario analysis of grey infrastructure strategies (doubling the conduit capacity) and the resulting total flood volume.

Scenario	Baseline	Grey1	Grey2	Grey3
Double capacity sections	-	J10 → Out1	J25 → Out1	J23 → J10
Drainage network graph				
Total flood volume (m^3)	1,505,233	967,052	898,032	1,498,943

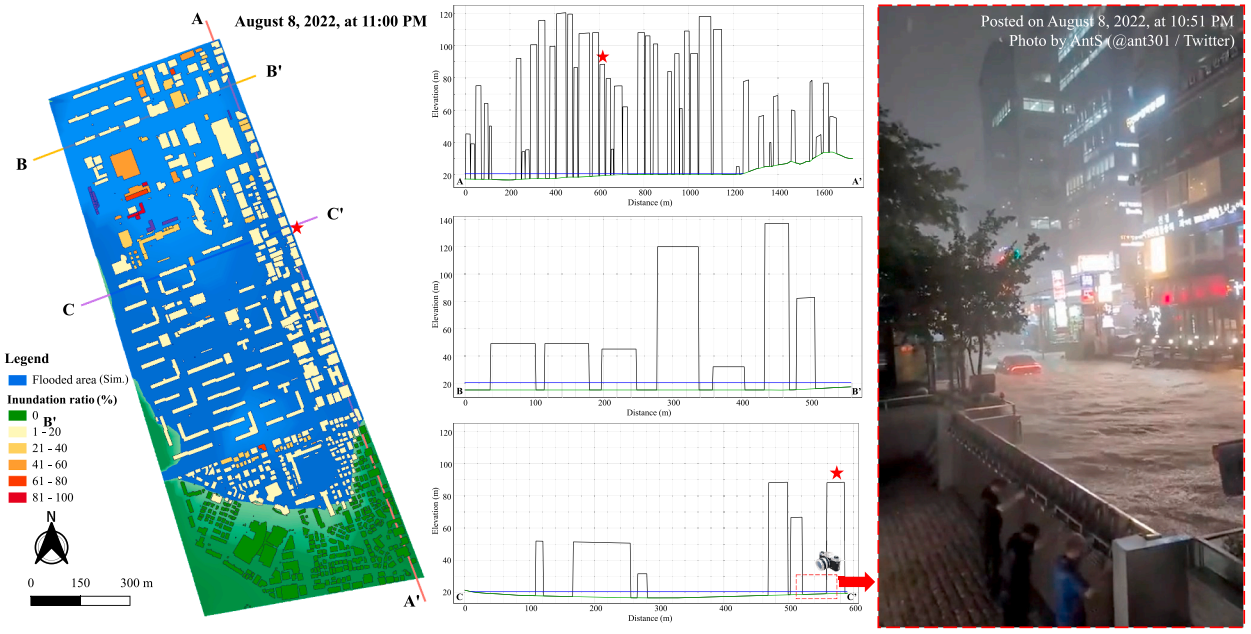


Fig. 11. Inundation ratio of each building and flood depth according to cross-sections A, B, and C simulated for 11 p.m. on August 8, 2022. The star symbol indicates an actual building that prevented flood damage through the installation of a flood barrier, as shown in the right picture (Photo by AntS (@ant301) / Twitter).

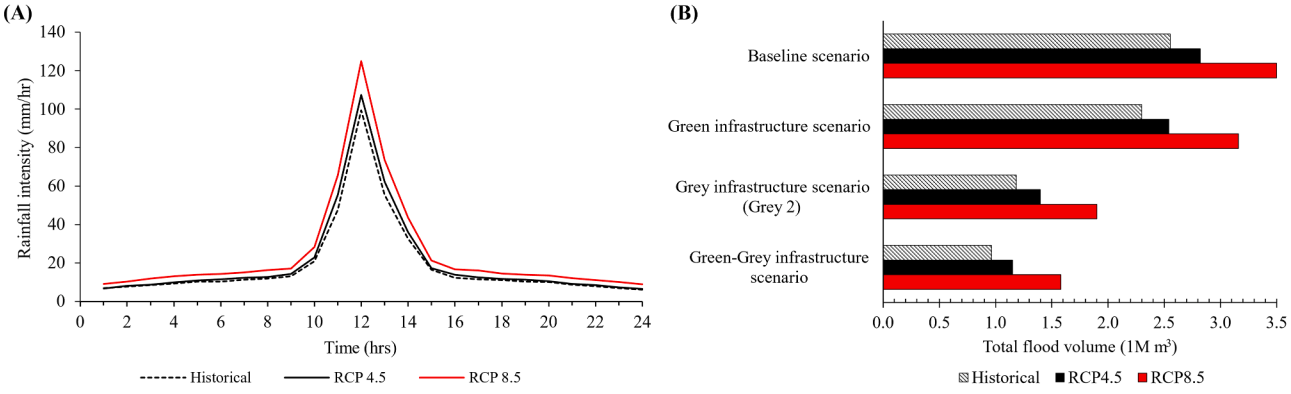


Fig. 12. (A) 24-hour design rainfalls for a 100-year recurrence interval according to climate change scenarios (Historical, RCP4.5, and RCP8.5) and (B) Flood mitigation effects of green and grey infrastructure under climate change scenarios.

(GRIMs) for Seoul (Hong et al., 2013). In particular, we utilized 24-hour design rainfall data for a 100-year return period in Seoul, incorporating historical data as well as the RCP4.5 and RCP8.5 scenarios (Fig. 12.A). To address the increasing risk of pluvial flooding caused by climate change, this pilot study investigated the best stormwater management practices for urban flood mitigation through coupled green- (Section 4.4.1) and grey infrastructure, specifically the Grey2 scenario (Section 4.4.2), as illustrated in Fig. 12.B. As a result, the flood mitigation effect of grey infrastructure averages 49.9% across all climate change scenarios, which is more effective in reducing flood volume compared to the average of 9.9% for green infrastructure. However, when implementing both green and grey infrastructure together, it has been found that the floodwater volume can be reduced by an average of 58.8% compared to the baseline scenario.

By the way, as it moves towards the RCP8.5 scenario, where high rainfall intensity is concentrated in short periods, the vulnerability to urban pluvial floods increases in all stormwater management strategies. This not only leads to an increase in the absolute amount of flood volume but also results in a decrease in the relative reduction rate of flood volume compared to the baseline. As demonstrated in Table 4, while green infrastructure exhibits weaker performance in mitigating flood volume, it is important to note that its relative reduction rate does not significantly diminish as climate change intensifies, especially when compared to grey infrastructure. This reflects the characteristic of green infrastructure to maintain similar performance even under weather extremes, supporting the rationale for integrating green infrastructure in conjunction with grey infrastructure to compensate for each other's weaknesses. Hence, it can be concluded that through coupled green-grey infrastructure, it is possible to reduce flood volumes by more than half for the study area, making it an effective strategy for eventually adapting to climate change and contributing to the creation of sustainable cities.

5. Discussion

5.1. Digital twins with bottom-up approach

In the face of the growing threat of pluvial flooding in urban environments, we raise the urgent need for an innovative solution that not only embraces modern technology but also actively involves citizen participation. To adaptively response to the spatial uncertainty associated with urban pluvial flooding, we need digital twin technology with a bottom-up approach that is capable of detailed, real-time analysis across the city scale. This approach encourages citizens, especially those most affected, to have a voice and actively engage in all stages, including the creation and analysis of urban pluvial flood models, and to contribute to decision-making, evaluation, and feedback processes. Incorporating citizens into this process enhances both the data reliability and the plausibility of the modeling (Oti et al., 2019), while also democratizing the decision-making process for flood risk management (Gharaibeh et al., 2021; Rodríguez-Bolívar et al., 2018). This represents an important paradigm shift: citizens can be active contributors, rather than just passive recipients of disaster information, leading to more sustainable and resilient communities in flood management (Bouzguenda et al., 2019).

The proposed framework highlights four essential components for participatory flood modeling and offers guidance on how various technologies should synergize and operate collaboratively. To recap, Open Source Software liberalizes access (Mahajan et al., 2022), Virtual Reality enhances situational awareness (Bakhtiari et al., 2023), Minimum Viable Product ensures real-time feedback (Xia et al., 2022), and Gamification facilitates engagement and quality (Ligorio et al., 2023). Each of these components contributes to making flood modeling more inclusive and actionable for both experts and the general public. This framework can help model a physical environment in real-time, allowing for rapid assessment of natural disaster situations. During emergencies such as floods, droughts, heat and cold waves, and environmental pollution, digital twins can be used to anticipate the trajectory and impact, enabling prompt and effective emergency responses.

Constructing these models involves coupling the stormwater and overland flow models, leveraging a 1D-RFSM model to enhance efficiency and streamline processes. In regions where dynamic observational data are scarce, utilizing multifaceted data sources and novel methods facilitates comprehensive model verification, empowering individuals and communities to devise and implement proactive and resilient stormwater management strategies. The final product of this model is a flood inundation map, and it is also possible to estimate the flood volume, flood duration, and peak flooding rate for each flooded node. Using a concrete case study in the Gangnam region of South Korea, the results showcase versatile applications to enhance urban flood resilience against climate change scenarios, encompassing both green and grey infrastructure along with collective action.

In terms of the future applicability of the model outputs, an online interactive platform or crowdsourcing archive for the participatory model is needed, as exemplified by the case of 'SWATShare' (Rajib et al., 2016). Firstly, the results can be utilized to create a 3D interactive flood risk map that displays inundation risks for individual buildings. Using this, an early warning and evacuation system can be established. Additionally, local authorities can use it for road traffic control in areas anticipated to be inundated, thus minimizing loss of life and property damage. Secondly, by integrating real-time data, this framework can aid in logistical planning and organizing resources during emergencies, such as rescue operations and disaster relief distribution (Zou et al., 2023). It facilitates collective decision-making by pinpointing the most accessible routes for relief delivery and identifying areas in urgent need for rescue. Lastly, to support collective action for flood control, this framework can also determine which heights and lengths of water barriers would divert surface runoff and influence flooding depths in flood-prone areas. This provides scientific evidence that can be used to advocate for necessary infrastructure improvement or to determine the required amount of human and material resources for collective action.

Given these points, digital twins offer a tool for a wide range of applications, especially in real-time monitoring and decision-making scenarios where there is a need to evaluate the potential impact of disaster mitigation strategies. This framework effectively articulates the importance for a bottom-up approach to supplement the traditional top-down strategies in urban flood management, emphasizing the community's role in addressing flood risks. As a caveat, the construction and maintenance costs for green infrastructure and grey infrastructure were not factored into this stylized exploratory model. The drainage network model in this study assumed a free outfall condition, and the conveyance capacity of the waterbodies was not considered when discharging runoff into them. Subsequent studies would need to incorporate these factors to facilitate more effective flood adaptation policies and collective action. By leveraging cutting edge technologies with knowledge sharing, the proposed framework offers a promising pathway to more resilient and adaptive urban communities in the face of climate change (Chang et al.,

Table 4
Simulated total flood volume of each stormwater management strategy under the 100-year rainfall with RCP4.5 and RCP8.5 climate change scenarios (The percentage in parentheses indicates the reduction rate of flood volume compared to the baseline).

Stormwater management strategy	Historical	RCP4.5	RCP8.5
Baseline scenario (No intervention) Total flood volume (m^3)	2,554,633	2,820,841	3,498,264
Green infrastructure scenario Total flood volume (m^3)	2,300,593 (-9.94%)	2,540,984 (-9.92%)	3,159,411 (-9.69%)
Grey infrastructure scenario (Grey 2) Total flood volume (m^3)	1,183,445 (-53.67%)	1,397,860 (-50.45%)	1,900,567 (-45.67%)
Green-Grey infrastructure scenario Total flood volume (m^3)	964,582 (-62.24%)	1,150,284 (-59.22%)	1,581,336 (-54.8%)

2021).

5.2. Participatory urban flood modeling and sustainable cities

Participatory urban flood modeling can contribute to achieving the SDG targets through each of its components (Fig. 13). However, when the four components work together in synergy with a digital twin model, they can have an even greater impact on achieving the SDGs.

Specifically, OSS can promote inclusive participation across all ages, genders, races, and socio-economic statuses (Target 10.2: Promote universal social, economic and political inclusion), leveraging its license-free, economically affordable nature and the collective contributions from individuals of diverse backgrounds (Rodríguez Bolívar et al., 2023). Moreover, OSS makes its source code and data openly and equally available to all, allowing anyone to conduct audits and verification freely. This approach boosts social capital and environmental equity, ensuring the public has access to information (Target 16.10: Ensure public access to information and protect fundamental freedoms). OSS exhibits flexible behavior through continuous model modifications and integration with other software in response to public demands. Consequently, this self-organizing feature of OSS advances participatory environmental modeling, which can be transferred and adapted in areas with restricted budgets and those vulnerable to climate-related disasters (Target 17.7: Promote and transfer sustainable technologies to developing countries).



Fig. 13. Components of participatory urban flood modeling and the Sustainable Development Goals. Abbreviations are as follows: OSS = open source software, VR = virtual reality, and MVP = minimum viable product (SDG icons by The Global Goals).

VR can serve as a tool that helps the general public more easily access, intuitively understand, and communicate disaster-related information (Target 9.C: Universal and affordable access to information and communications technology). By offering an immersive experience that mirrors the real world, it enhances situational awareness about disaster situations, thereby increasing the adaptive capacity of individuals and communities to climate-related disasters. Additionally, citizens and experts can share and disseminate accumulated knowledge through repeated scenario analysis for decision-making on best management practices (Target 13.3: Improve education, awareness-raising, and human and institutional capacity). VR, combined with engaging technologies like GIS, the Internet of Things, and the metaverse, can induce high usability and active citizen participation, leading to collective actions such as the installation of temporary flood barriers in scientifically informed specifications and locations (Target 17.6: Knowledge sharing and cooperation for access to science, technology, and innovation).

For rapid flood assessment and early warning, a simple and rapid MVP is an technically important attribute of participatory urban flood modeling (Target 3.D: Improve early warning systems for global health risks). Given the technical and budgetary constraints on hardware for real-time emergency response systems, focusing on streamlining software operations in participatory flood modeling offers the most effective approach. This involves reducing computational power while still being able to reliably represent core physical processes with an MVP (Target 9.5: Enhance scientific research and upgrade the technological capabilities of industrial sectors). Integrating participatory modeling enables the creation of easily adaptable and stylized archetype collections for each Level of Detail of digital twins, specifically tailored to areas requiring disaster response (Target 17.8: Operationalize the technology bank and science, technology, and innovation capacity for least developed countries).

VGI can be used to obtain a large amount of high-quality, real-time data on urban pluvial floods at a low cost for flood model validation. However, to keep VGI responsive and reliable, gamification methods that combine reward (i.e., non-monetary payoff) and sanction systems need to be incorporated (Target 17.18: Enhance availability of high-quality, timely and reliable data). Gamification also supports and strengthens local community engagement, becoming a foundation in sustainable stormwater management by providing implicit knowledge, the community's memories, and past experiences with floods (Target 6.B: Support local engagement in water and sanitation management). When citizens are motivated by gamification to contribute to any process of participatory modeling, they learn how individual voluntary cooperation can affect community disaster resilience. This hands-on approach not only fosters a deeper understanding of collective action but also serves an educational role in promoting the SDGs as global citizens (Ligorio et al., 2023; Meyer et al., 2018) (Target 4.7: Education for sustainable development and global citizenship).

When these four components of participatory urban flood modeling work together through digital twins with a bottom-up approach, they can significantly contribute to achieving the following six SDG targets: Target 1.5, 6.5, 11.5, 11.B, 13.1, and 16.7.

Participatory urban flood modeling empowers communities by involving them throughout the entire process. This approach not only enables them to identify pluvial flood risks but also to collectively devise, verify, and implement strategies that mitigate the social-

ecological impact of flood damages (Target 1.5: Build resilience to environmental, economic and social disasters). Enhanced identification and comprehensive planning against flood risks through participatory modeling are crucial for minimizing both casualties and financial losses. Specifically, the development of flood mitigation strategies and early warning systems via this bottom-up approach offers substantial protection to vulnerable populations in urban areas (Target 11.5: Reduce the adverse effects of natural disasters). By creating a collaborative platform, participatory urban flood modeling fosters open communication and knowledge sharing among various stakeholders involved in the decision-making process (Target 16.7: Ensure responsive, inclusive and representative decision-making). This can lead to the development of more integrated stormwater management practices that prioritize social equity, economic efficiency, and ecosystem sustainability (Biswas, 2004; Rahaman & Varis, 2005) (Target 6.5: Implement integrated water resources management). Moreover, this modeling approach deepens the collective understanding of flood risks and challenges in stormwater management in urban environments, aligning with the Sendai Framework for Disaster Risk Reduction (Target 11.B: Implement policies for inclusion, resource efficiency and disaster risk reduction). Hence, the interplay between citizen participation and digital twins serves a pivotal role in bolstering community disaster resilience by enhancing flood risk diagnosis, developing effective flood mitigation strategies, strengthening community engagement, and democratizing decision-making (Target 13.1: Strengthen resilience and adaptive capacity to climate related disasters).

Overall, participatory urban flood modeling can ultimately contribute to the achievement of the following 10 SDGs: Goal 1, 3, 4, 6, 9, 10, 11, 13, 16, and 17.

6. Conclusion

To address the escalating threats of urban pluvial flooding, it is required to leverage digital twin technology with a bottom-up approach, enhancing disaster preparedness and response at both individual and community levels. This paper presents a participatory modeling framework by highlighting four pivotal components — open source software, virtual reality, a minimum viable product, and gamification — embodying the essence of the digital twin era. Firstly, our novel framework encompasses key processes of input data preparation, hydrological model construction, model verification, and experimentation. Secondly, this framework was applied in the Gangnam region of Seoul, South Korea, by coupling a one-dimensional stormwater model with a rapid flood spreading model. The case study demonstrated that the model achieved plausible accuracy when evaluated using double validation, which combines quantitative (i.e., Lee-Sallee index = 0.89, NSE = 0.52, and RSR = 0.69) and qualitative (i.e., visual comparison with social media images) verification methods. Through participatory modeling, citizens can examine the pluvial flood mitigation effects of both green and grey infrastructure strategies, as well as collective action, under various climate change scenarios. As a result, grey infrastructure more effectively reduces flood volume than green infrastructure (49.9% vs. 9.9% on average) across various climate change scenarios, but combining them can maximize the flood reduction (58.8%). So as to ensure consistent and reliable performance amidst climate change uncertainty, adopting coupled green-grey infrastructure and collective flood mitigation strategies is crucial for the study area. This framework promotes a paradigm shift by transforming citizens' roles from passive receivers to proactive contributors in making cities and society more sustainable against climate-related disasters. Ultimately, this framework embarks on paving a pathway towards fostering more informed decision-making, armed with the digital twin technology and community engagement, ready to confront extreme weather-induced challenges.

CRediT authorship contribution statement

Samuel Park: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jaekyoung Kim:** Visualization, Validation, Software, Project administration, Formal analysis, Data curation. **Yejin Kim:** Visualization, Validation, Software, Project administration, Formal analysis, Data curation. **Junsuk Kang:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data access statement The research data supporting this publication are openly available at: https://doi.org/10.17632/54pv8f6ynr.3.

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Supplementary materials

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